

CS-AML

UX Specification

Version 0.1

User experience requirements for the Civil Society Anti-Money Laundering / Financial Intelligence Platform

Status

Normative UX baseline for MVP 0.1. This document defines interaction behaviour, user-facing semantics, task flow, feedback, safety, accessibility, and usability requirements. It intentionally does not define final information architecture, taxonomy, sitemap, or object hierarchy; those belong to the separate CS-AML Information Architecture Specification.

Document	Relationship
PRD v0.1	Defines product outcomes and MVP scope.
SRS v0.1	Defines testable software requirements.
MVP Engineering Breakdown v0.1	Defines implementation epics and stories.
Technical Stack & Repository v0.1	Defines technical implementation baseline.
This UX Specification	Defines user interaction behaviour and experience quality.
Future IA Specification	Will define content/object organisation, taxonomy, navigation model, labels, hierarchy, and findability.

CS-AML Framework Suite • 2026

1. Purpose and Scope

The purpose of this specification is to make the CS-AML investigation experience safe, intelligible, reviewable, and efficient for civil-society investigators working with sensitive financial intelligence. It translates product and analytical principles into interaction requirements that designers and frontend engineers can implement and test.

UX axiom

The interface SHALL preserve analytical uncertainty rather than visually collapse it. Facts, claims, inferences, candidate matches, hypotheses, reconstructed value flows, and final assessments must remain perceptibly different throughout the user journey.

1.1 In scope

- Interaction models and user task flows.
- User-facing semantics for provenance, confidence, status, uncertainty, sensitivity, and review state.
- Data-entry behaviour, validation, confirmation, undo, conflict handling, and error recovery.
- Evidence handling, entity-resolution, graph, timeline, value-flow, typology, hypothesis, assessment, review, and dissemination experiences.
- Accessibility, keyboard operation, desktop layouts, responsive constraints, loading/empty/error states, and usability verification.
- User-facing behaviour for permission boundaries, protected sources, audit visibility, and AI-assisted features.

1.2 Explicitly out of scope

- Final sitemap or menu tree.
- Canonical navigation taxonomy and object hierarchy.
- Global naming/label taxonomy beyond interaction-specific terms.
- Content inventory and findability model.
- Final URL structure and route taxonomy.
- Cross-domain information architecture decisions.

Those topics SHALL be defined in the separate CS-AML Information Architecture Specification so that UX and IA can be owned by different engineering/design workstreams without ambiguity.

2. Target Users and Usage Context

Role	Primary UX needs	High-risk moments
Investigator	Fast evidence capture, structured analysis, clear provenance, low-friction linking.	Uploading evidence, creating adverse relationships, changing scope, exporting.
Case Owner	Case status, gate readiness, risk visibility, approval context.	Gate approval, dissemination, closure/reopen.

Reviewer	Independent reconstruction of reasoning, contradictions, source quality and confidence.	Approving assessments/products, challenging identity resolution.
Evidence Custodian	Original preservation, hash/integrity, derivatives, provenance.	Replacement, mismatch, disposition.
Data Steward	Entity identity quality, merge/unmerge, vocabulary consistency.	High-impact merge, conflicting identifiers.
Source Handler	Protected source compartment and safe evidence handoff.	Identity reveal, export, search leakage.
Administrator/Auditor	Policy, retention, access and audit observability.	Role changes, retention execution, incident review.

2.1 Operating conditions

- Desktop-first investigation work, frequently involving multiple documents and long analytical sessions.
- Potentially slow or unreliable networks; users must not lose work silently.
- Cases may involve personally sensitive or reputationally harmful information.
- Analysts may work across multiple languages and jurisdictions.
- Users may be technically sophisticated but should not need to understand database or graph internals.
- Reviewers may enter a case late and must quickly understand what is known, inferred, disputed, and unknown.

3. UX Goals and Quality Attributes

Goal	Meaning	UX success signal
U1 Traceable	Users can move backward from judgement to evidence without losing context.	Reviewer reaches supporting evidence within a small number of intentional interactions.
U2 Uncertainty-aware	The interface makes fact/inference/hypothesis states perceptible.	Users can correctly identify status in usability testing.
U3 Reversible	High-impact actions can be reviewed, undone, superseded, or corrected.	No irreversible merge/edit occurs without explicit confirmation and recovery path.
U4 Safe-by-default	Sensitive actions require context and authorization, not accidental clicks.	Exports/reveals cannot occur unintentionally.
U5 Efficient	Repeated investigative work avoids needless re-entry and context switching.	Common workflows complete without duplicate data entry.
U6 Reviewable	A reviewer can understand analytical	Reviewer can inspect support,

	reasoning independently.	contradiction, assumptions and gaps.
U7 Accessible	Core workflows are keyboard and assistive-technology usable.	Critical operations do not rely on color, pointer precision, or hover only.
U8 Calm	The system supports deliberate judgement rather than alert overload.	No gratuitous urgency or suspicion scoring in default interface.

4. Core UX Principles

UX-P01 Evidence before assertion

When a user creates a material fact, relationship, asset attribution, value flow, indicator or assessment, the interface SHOULD make the evidence link visible and easy to add.

UX-P02 Status is visible

Candidate, confirmed, disputed, rejected, superseded, direct, reconstructed, hypothetical, draft, approved and restricted states SHALL never be hidden only in metadata.

UX-P03 No silent promotion

A candidate or AI suggestion SHALL NOT visually or functionally become a canonical fact without explicit user action.

UX-P04 No accusation by styling

Red color, warning icons, threat metaphors, or risk-like visual emphasis SHALL NOT imply criminality merely because an entity appears in an investigation.

UX-P05 Progressive disclosure

Advanced provenance, audit and metadata are available without overwhelming the primary task.

UX-P06 Reversibility

Merge, classification change, dissemination, correction, disposition and other high-impact actions SHALL expose consequences before commitment.

UX-P07 Context preservation

Opening evidence, graph edges, timeline events or hypotheses SHOULD preserve case context and allow return without losing state.

UX-P08 Explicit unknowns

Unknown, unavailable and not-yet-assessed are distinct from zero, none, false or cleared.

UX-P09 Reviewer independence

Review surfaces SHOULD support challenge and contradiction, not merely sign-off.

UX-P10 Permission dignity

Denied access SHALL be handled without leaking sensitive object existence or encouraging users to bypass controls.

5. Interaction Model and Analytical State Semantics

5.1 User-visible analytical states

Concept	Required visual semantics	Prohibited simplification
Claim	Attributed statement from a source; not yet accepted as fact.	Displaying as verified fact.
Fact	Verified/time-bounded proposition with evidence.	Hiding provenance.
Candidate entity match	Possible identity equivalence requiring review.	Auto-merge appearance.
Relationship	Typed connection with evidence/status/time.	Graph edge with no provenance status.
Indicator	Observed pattern relevant to typology.	Label as suspicious/proven crime by default.
Hypothesis	Testable explanation with support and contradiction.	Present as conclusion.
Assessment	Analyst judgement with confidence and gaps.	Present confidence as mathematical probability unless validated.
Value flow class	DIRECT / DOCUMENTED / RECONSTRUCTED / HYPOTHETICAL.	One undifferentiated arrow style.
Intelligence product	Versioned product with review/approval state.	Overwrite prior approved version.

5.2 Status presentation rule

Every high-impact object SHALL expose its status near the object title or primary identifier, using text labels plus secondary visual encoding. Color MAY reinforce status but SHALL NOT be the only carrier of meaning.

6. End-to-End UX Journey

1. Open or create a governed case with a clear investigation question and scope.
2. Register a source and preserve original evidence.
3. Create extracts or claims while retaining provenance.
4. Create or resolve entities and link identifiers without premature merge.
5. Build evidence-backed relationships, ownership/control assertions and assets.

6. Place events on a chronology and reconstruct value flows with explicit class.
7. Compare observations to typologies and capture counter-indicators.
8. Test competing hypotheses, document gaps and produce a confidence-rated assessment.
9. Generate an intelligence product with an evidence index.
10. Complete independent review, resolve changes, approve dissemination and record sharing.

Journey rule

The user **SHOULD** be able to move forward and backward through this chain without losing provenance or being forced to duplicate canonical information. The UX **SHALL** support iteration: investigation is not a one-way wizard.

7. Case and Workflow Experience

7.1 Case creation

Requirement ID	UX requirement	Acceptance
UXR-CASE-001	Case creation SHALL make owner, purpose, investigation question and classification understandable before activation.	User cannot accidentally activate an incomplete case; missing fields explain why.
UXR-CASE-002	Investigation question entry SHALL encourage neutral wording and avoid guilt-presuming templates.	Prompt/example text uses question framing, not accusation framing.
UXR-CASE-003	Material scope changes SHALL show what changes and require rationale.	User can compare current and proposed scope before saving.
UXR-CASE-004	Lifecycle gate status SHALL be visible at the point of blocked actions.	Blocked action explains gate requirement and authorized next step.
UXR-CASE-005	Task completion SHALL not hide historical ownership or prior state.	Closed task retains owner, timestamps and linked objects.

7.2 Gate and approval interactions

- Gate approval **SHOULD** summarize the decision context: requested action, case classification, unresolved gaps, risk conditions, and reviewer independence requirement.
- Approval/rejection **SHALL** require an intentional action and **SHALL** not share a primary button style with navigation controls.
- When self-approval is prohibited, the UI **SHALL** explain that a different authorized reviewer is required rather than presenting a generic error.

8. Source, Evidence and Document Experience

8.1 Evidence ingestion

ID	UX requirement	Acceptance
UXR-EVD-001	Upload experience SHALL separate original evidence from analyst-created derivatives.	Original and derivative have visibly different labels and actions.
UXR-EVD-002	After upload, integrity/hash status SHOULD be visible without requiring a technical console.	User can see success, pending or mismatch state.
UXR-EVD-003	Failed upload or scan SHALL never look like a successfully preserved evidence item.	Incomplete items are clearly marked and retryable.
UXR-EVD-004	Evidence extract creation SHALL retain parent and precise location.	User can return from extract to parent location.
UXR-EVD-005	Reliability and information credibility SHALL be entered separately.	One control cannot overwrite or substitute for the other.
UXR-EVD-006	Mutable web sources SHOULD expose whether an archived copy exists.	Missing archive is distinguishable from archived status.

8.2 Document reading interaction

- The reading view SHOULD allow side-by-side or quickly switchable evidence context and analytical notes without requiring users to lose their place.
- Citations/extracts SHOULD be creatable from the reader with visible source/page context.
- Derived OCR text SHOULD be labelled as extracted text and not replace the visual/original evidence representation.
- Copying text from evidence SHOULD preserve or offer citation metadata where practical.

9. Entity Resolution UX

High-impact design principle

Entity resolution is a decision, not a convenience feature. Candidate similarity SHALL be presented as evidence for review, not as proof of sameness.

ID	UX requirement	Acceptance
UXR-ENT-001	Candidate match view SHALL present matching and conflicting attributes side by side.	Reviewer can see both evidence for and against merge.

UXR-ENT-002	Common-name similarity SHALL not receive visual treatment equivalent to a confirmed identifier match.	Similarity rationale is explicit.
UXR-ENT-003	Merge confirmation SHALL summarize affected aliases, identifiers, relationships, cases and provenance.	User sees consequence preview.
UXR-ENT-004	Unmerge/reversal path SHALL be discoverable to authorized users.	No hidden support-only reversal for MVP.
UXR-ENT-005	Entity status such as candidate, probable, confirmed, disputed or unresolved SHALL be visible in graph and detail views.	Status is text-visible, not color-only.

10. Relationship, Ownership and Asset UX

10.1 Relationship creation

- Relationship creation SHALL require explicit relationship type selection.
- The interface SHALL distinguish legal ownership, beneficial ownership, control, use and association.
- Unknown percentage SHALL be representable without defaulting to 0%.
- Relationship status, valid time and evidence SHALL be reviewable from any visual edge representation.

10.2 Asset attribution

- The interface SHALL distinguish “owned by”, “controlled by”, “used by”, and “associated with”.
- Estimated valuation SHALL visibly show date, currency, source and uncertainty.
- A material asset attribution without evidence SHOULD trigger a visible incomplete-state cue before assessment use.

11. Timeline UX

ID	UX requirement	Acceptance
UXR-TIM-001	Timeline SHALL preserve date precision.	Year-only/month-only events are not rendered as exact dates.
UXR-TIM-002	Events SHALL link back to evidence and canonical event details.	User can inspect basis without losing timeline context.
UXR-TIM-003	Filters SHALL not silently change analytical meaning.	Active filters are always visible and clearable.
UXR-TIM-004	Temporal correlation SHALL not be worded as causation.	Default labels use neutral temporal language.

12. Follow-the-Value UX

12.1 Flow classes

Class	Required user-facing meaning	Minimum visual cue
DIRECT	Supported by direct transactional evidence.	Text class label + solid relationship semantics.
DOCUMENTED	Movement/value relationship supported by documentary evidence but not necessarily a direct account transaction.	Text label distinct from DIRECT.
RECONSTRUCTED	Analytical reconstruction from multiple evidence points.	Text label + non-solid/differentiated line semantics.
HYPOTHETICAL	Proposed flow used for testing a hypothesis.	Text label + clearly tentative visual treatment.

Color SHALL NOT be the sole method for distinguishing classes. Legend/labels SHALL survive screenshot, print and export.

12.2 Flow builder

- Users SHOULD build flows leg by leg, with evidence and class attached per leg.
- Unknown amount, approximate amount, range and exact amount SHALL be distinct input states.
- The interface SHALL not coerce unknown amount to zero.
- Deleting a visual flow leg SHALL not delete underlying evidence.
- Before a reconstructed or hypothetical flow is used in an assessment, the interface SHOULD show its class and confidence in the assessment context.

13. Typology and Indicator UX

ID	UX requirement	Acceptance
UXR-TYP-001	Typology catalogue SHALL display mechanism, observables, counter-indicators, false positives and version.	User can inspect cautionary context before matching.
UXR-TYP-002	Indicator capture SHALL require neutral proposition wording and linkable evidence.	Indicator is not labelled “crime detected”.
UXR-TYP-003	Typology match SHALL expose observed indicators and counter-indicators together.	Counterevidence is not hidden in secondary tab only.
UXR-TYP-004	Consistency level SHALL require analyst rationale.	System cannot silently elevate match level.
UXR-TYP-005	One weak indicator SHALL not generate	Visual emphasis reflects evidence quality,

	strong visual warning by default.	not mere presence.
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14. Hypothesis and Assessment UX

14.1 Competing hypotheses

Reasoning UX

The hypothesis workspace **SHOULD** make it easy to record evidence that weakens the analyst's preferred explanation. The interaction should reward completeness, not confirmation.

- Multiple hypotheses **SHOULD** be visible in parallel comparison where screen space allows.
- Supporting and contradicting evidence **SHALL** be visually separable without implying numerical scoring.
- Rejected hypotheses **SHALL** remain accessible in history.
- Assumptions and intelligence gaps **SHOULD** be shown near the hypothesis they affect.

14.2 Assessment drafting

ID	UX requirement	Acceptance
UXR-ASM-001	Assessment authoring SHALL separate judgement, confidence, basis, assumptions, alternatives and gaps.	A single narrative box is insufficient for material assessment.
UXR-ASM-002	Confidence selection SHALL require rationale and SHOULD show descriptive guidance.	User cannot save material assessment with confidence but no basis.
UXR-ASM-003	Assessment SHALL expose backward trace to supporting evidence.	Reviewer can traverse to provenance.
UXR-ASM-004	Unresolved gaps SHALL remain visible when product is drafted.	Drafting report does not hide gaps.
UXR-ASM-005	System SHALL avoid probability-like visual precision unless model is validated.	No default 0-100 "guilt/risk score".

15. Search and Investigation Graph UX

15.1 Search interaction

- Search results **SHALL** never reveal unauthorized object titles, snippets, counts or facets.
- Filters and case scope **SHALL** remain visible while results are shown.
- Search result type, status and case context **SHOULD** be distinguishable.

- Search is a discovery tool; the interface SHALL not imply that result ranking equals analytical significance.

15.2 Graph interaction

ID	UX requirement	Acceptance
UXR-GRF-001	Every edge SHALL expose relationship type, status and provenance access.	No anonymous decorative edges for material relationships.
UXR-GRF-002	Graph SHALL distinguish entity status and flow class without color-only encoding.	Text/shape/line semantics are available.
UXR-GRF-003	Expanding a node SHALL not imply discovered entities are analytically relevant.	Expansion is neutral and reviewable.
UXR-GRF-004	Filtering SHALL show active filter state and allow reset.	User cannot unknowingly interpret a partial graph as complete.
UXR-GRF-005	Graph layout changes SHALL not change canonical data.	Repositioning nodes is presentation-only.

16. Review Experience

16.1 Independent review workspace

- Reviewer SHOULD see the product version, author, case scope, assessment confidence, unresolved gaps, source/evidence index, and review status in one coherent review context.
- Reviewer SHALL be able to comment or request change without editing the author's approved history directly.
- The interface SHOULD support challenge prompts covering identity, provenance, alternative explanations, confidence and potential harm.
- Where independence is required, the author SHALL not see an enabled self-approve control.

16.2 Review resolution

State	UX behaviour
Draft	Editable by authorized author; clearly not approved.
In review	Version is review-targeted; major edits create or update draft according to product rules.
Changes requested	Reviewer comments and required changes are visible and actionable.
Approved	Approved version is frozen; edits require new version.

Superseded	Prior version remains readable with link to current version.
Retracted	Status and reason remain visible; record is not silently deleted.

17. Dissemination and Export UX

Safety gate

External dissemination is a consequential action. The UX SHALL make recipient, purpose, classification, version, included objects, minimisation/redaction state, and handling restrictions visible before final export.

ID	UX requirement	Acceptance
UXR-DIS-001	External export SHALL be blocked until required approval exists.	No hidden bypass through alternate export button.
UXR-DIS-002	Export preview SHALL show what will be included and excluded.	User can inspect manifest before generation.
UXR-DIS-003	Restricted/protected source identity SHALL never be included by default.	Explicit policy-controlled inclusion only if authorized.
UXR-DIS-004	Sharing completion SHALL show recipient, version, time and restrictions.	Sharing log confirmation is visible after action.
UXR-DIS-005	Correction/supersession state SHALL appear on later exports.	Recipient-facing document identifies current/superseded status.

18. Permission and Protected-Source UX

18.1 Permission feedback

- Denied access SHOULD use non-disclosing language such as “You do not have access to this resource” rather than confirming sensitive object identity.
- If access can legitimately be requested, the interface MAY provide a controlled request path without revealing protected metadata.
- Permission state changes SHOULD be reflected promptly without requiring users to infer stale access.

18.2 Protected source interactions

- Protected source identity SHALL be visually and functionally compartmentalized from routine source-derived evidence.
- Routine analyst views SHALL reference a protected source token/alias rather than identity.

- Reveal actions, where permitted, SHALL require explicit intent and SHOULD be logged as high-sensitivity audit events.
- Protected identity SHALL not appear in ordinary search, graph labels, exports, autocomplete, browser title text, or routine notifications.

19. Form and Data Entry Standards

Pattern	Requirement
Required fields	Required vs optional SHALL be explicit before submit.
Unknown	Provide “Unknown / Not established” where analytically valid; do not force false precision.
Date precision	Support exact date, month, year, range and unknown.
Money/value	Support exact, approximate, minimum, maximum, range and unknown; preserve currency.
Confidence	Use controlled descriptive scale with rationale; avoid pseudo-precision.
Autosave	Long-form analytical drafting SHOULD protect against accidental loss while preserving version/audit semantics.
Validation	Validation messages SHALL state what is wrong and how to resolve it.
Destructive action	Use consequence preview and explicit confirmation; typed confirmation reserved for exceptionally high-impact actions, not routine use.
Cancel/back	Leaving a dirty form SHALL warn unless safe autosave/versioning exists.

20. System Feedback and State Handling

State	Required UX behaviour
Loading	Show activity without implying completion; long jobs SHOULD expose progress/state where available.
Empty	Explain what the area represents and the next legitimate action; avoid making absence look like “cleared/no risk”.
Partial	Clearly label partial or filtered results.
Error	Preserve user input where safe and provide recovery action.

Offline/interrupted	Warn before losing work; retry idempotently where applicable.
Background job	Show queued/running/completed/failed; prevent duplicate submission.
Conflict	When another user/version changed the object, show comparison or safe reload path rather than silent overwrite.
Permission changed	Terminate or refresh affected actions; do not continue with stale authorization.
Integrity mismatch	Use high-severity but factual language; do not allow normal analyst dismissal without authorized process.

21. Undo, Correction and Reversibility

- Routine edits SHOULD support undo or version recovery where feasible.
- Entity merge MUST support authorized unmerge.
- Approved products MUST use correction/supersession, not destructive overwrite.
- Relationship deletion SHOULD preserve audit history; where possible, status “withdrawn/rejected/superseded” is preferable to erasure.
- Export/share actions cannot be undone externally; the UX SHALL explicitly communicate this before dissemination.

22. Accessibility Requirements

ID	Requirement
UXR-A11Y-001	Core workflows SHALL be operable with keyboard only.
UXR-A11Y-002	Focus indicators SHALL remain visible and logical.
UXR-A11Y-003	Color SHALL NOT be the sole carrier of flow class, status, warning, evidence quality or confidence.
UXR-A11Y-004	Interactive graph/timeline SHALL have accessible alternative representations such as lists/tables.
UXR-A11Y-005	Form controls SHALL have programmatic labels and useful validation text.
UXR-A11Y-006	Text and UI contrast SHOULD meet WCAG 2.2 AA targets.
UXR-A11Y-007	Critical confirmation dialogs SHALL not trap or unpredictably move focus.
UXR-A11Y-008	Hover-only information SHALL also be available by

	focus/click or persistent text.
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23. Desktop, Responsive and Multi-Panel Behaviour

MVP is desktop-first. The interface SHOULD optimize for 1280px and wider investigation workspaces while remaining usable on smaller laptop displays. Mobile MAY support read-only or limited review flows later but is not a primary MVP target.

- Multi-panel layouts SHOULD collapse gracefully rather than compress critical text below readable width.
- Evidence/analysis side panels SHOULD remember reasonable user state within a session.
- Primary actions SHALL remain discoverable when a detail panel is open.
- Graph and timeline views SHOULD provide full-screen focus mode without severing provenance access.

24. Visual Semantics and Design System Behaviour

24.1 Semantic design tokens

Semantic intent	Use	Rule
Neutral	Most entities, facts, navigation and informational surfaces.	Default visual state.
Informational	Derived/secondary context, system notices.	Must not imply suspicion.
Caution	Incomplete provenance, unresolved contradiction, high-impact action.	Use factual wording.
Critical	Integrity mismatch, security failure, blocked unsafe action.	Reserved for actionable system/safety conditions.
Positive	Completed verification, approved review, successful integrity check.	Do not use to imply “innocent/cleared”.

24.2 Prohibited visual patterns

- Crime-style red highlighting for a person/company merely because it is a subject or graph hub.
- “Risk gauge” or “guilt score” without validated model and explicit approved use.
- Flashing/pulsing alerts for analytical indicators.
- Graph size encoding that users could mistake as proof of importance without clear explanation.
- Hidden uncertainty revealed only on hover.

25. Notifications and Attention Management

- Notifications SHOULD be limited to actionable events: assignment, review request, approval/rejection, integrity/security event, failed processing, or required gate action.
- The system SHOULD avoid generic “suspicious activity” alerts as an attention mechanism.
- Sensitive case details SHALL be minimized in email/push notifications.
- Users SHOULD be able to distinguish urgent security/integrity events from routine workflow updates.

26. Collaboration UX

- Concurrent work SHOULD show object version or recent-change awareness where collisions are plausible.
- Comments/review notes SHOULD be attributable and timestamped.
- Users SHOULD not assume another analyst has read a comment merely because it is present; acknowledgement state MAY be used where useful.
- Collaborative editing SHALL not erase individual authorship of material analytical changes.

27. AI-Assisted UX Guardrails

AI interaction principle

AI output is a suggestion layer. It SHALL look and behave like generated/derived material until a human explicitly verifies or promotes the relevant proposition.

ID	UX requirement	Acceptance
UXR-AI-001	AI-generated text/candidate SHALL be labelled generated and include model/tool context where material.	User can distinguish generated output from evidence.
UXR-AI-002	AI suggestions SHALL require review before becoming canonical entity/relationship/fact.	No one-click silent promotion.
UXR-AI-003	AI answers SHOULD cite authorized canonical objects/evidence.	Unsupported answer is visibly limited.
UXR-AI-004	AI SHALL not present guilt determination, dissemination approval or identity merge as an autonomous decision.	Consequential action remains human-controlled.
UXR-AI-005	Protected-source identity SHALL be excluded from AI context unless explicitly authorized by policy.	Default context builder omits compartmented identity.

28. Onboarding and Learnability

- First-use guidance SHOULD explain the analytical chain and the difference between evidence, fact, indicator, hypothesis and assessment.
- Inline help SHOULD be contextual and concise; methodology depth may link to framework guidance rather than crowd the interface.
- Dangerous or uncommon actions such as merge, unmerge, dissemination and source reveal SHOULD include just-in-time guidance.
- Terminology SHALL remain consistent across forms, review states and exports.

29. Usability Testing and UX Verification

29.1 Required test scenarios

Scenario	Participant goal	Pass condition
UT-01 Case start	Create governed case and understand why activation is blocked/incomplete.	User completes without moderator explanation.
UT-02 Evidence trace	Create evidence extract and later return from assessment to exact evidence.	Provenance path succeeds.
UT-03 Entity resolution	Review candidate pair with conflicting attributes and decline/merge appropriately.	User understands candidate != confirmed.
UT-04 Value flow	Build mixed direct/reconstructed flow and interpret exported result.	Class distinction understood without color reliance.
UT-05 Hypothesis	Record support and contradiction for two hypotheses.	User can explain difference between hypothesis and assessment.
UT-06 Peer review	Reviewer finds gaps, evidence and alternatives and requests changes.	Independent review completes without author guidance.
UT-07 Dissemination	User attempts export before approval, then completes approved minimized export.	Unsafe path blocked; approved path understandable.
UT-08 Protected source	Routine analyst uses derived evidence but cannot discover source identity.	No leakage via search/export/labels.

29.2 UX success measures

- Task success rate for required scenarios.
- Critical error rate (e.g., mistaken merge, wrong flow class, accidental export attempt).
- Time-to-evidence trace for reviewer.
- Percentage of users who correctly identify fact vs hypothesis vs assessment states.

- Accessibility defects in critical workflow.
- User-reported confidence in understanding uncertainty and provenance.
- Number of support interventions needed for first end-to-end pilot.

30. UX Analytics and Privacy

Product analytics SHALL be privacy-preserving and SHALL NOT capture evidence content, protected identities, case narrative, document text, search terms, or graph payloads unless explicitly approved for a specific controlled research purpose.

- Prefer event-level telemetry such as screen/action type, duration, validation failure category and feature usage.
- Telemetry payloads SHOULD use opaque object identifiers where needed and SHOULD avoid sensitive labels.
- Session replay tools SHALL be disabled by default for sensitive investigation environments.
- UX research exports SHALL be minimized and reviewed before leaving the protected environment.

31. UX Requirement Traceability

UX family	Primary upstream source	Engineering workstream
UXR-CASE	PRD/SRS FR-CASE; E2	Case/workflow frontend
UXR-EVD	FR-EVD/DOC; E3	Evidence/document frontend
UXR-ENT	FR-ENT; E4	Entity resolution UI
UXR-TIM	FR-TIM; E5	Timeline UI
UXR-TYP	FR-TYP; E6	Analytical reasoning UI
UXR-ASM	FR-ASM; E6	Assessment UI
UXR-GRF	FR-GRF; E7	Graph/search UI
UXR-DIS	FR-PRD/REV/DIS; E8	Product/review/export UI
UXR-A11Y	SRS UX/NFR	Cross-cutting frontend
UXR-AI	SRS AI	Future assistive/AI layer

32. UX Definition of Done

1. Interaction satisfies applicable UXR requirements and SRS acceptance criteria.
2. Loading, empty, partial, error, conflict and permission-denied states are designed, not left implicit.
3. Keyboard navigation and focus behaviour are verified for critical workflows.

4. Color is not the sole carrier for analytical state, value-flow class or warnings.
5. Provenance and uncertainty remain visible at consequential decision points.
6. High-impact action has consequence preview, authorization handling and audit-aware confirmation.
7. Usability test or design review covers the relevant critical scenario.
8. No interaction introduces terminology or hierarchy that conflicts with the future IA specification without an explicit cross-document decision.

33. UX-IA Handoff Contract

Separation of ownership

UX owns interaction behaviour. IA owns organisation and findability. Both specifications SHALL share the same canonical domain vocabulary and resolve conflicts through explicit architecture/product decisions rather than local UI workarounds.

UX team owns	IA team owns	Joint decision
Task flows and interaction states	Sitemap/navigation model	Where a task enters/exits global structure
Forms, validation, feedback, undo	Object grouping and hierarchy	Object labels visible to users
Graph/timeline interaction behaviour	Content/object taxonomy	Cross-object discoverability
Review/dissemination interaction	Global navigation and findability	Role-specific entry points
Accessibility and interaction semantics	Label system and information scent	Terminology consistency
Loading/error/permission states	Route/content organisation	Deep-link and context preservation behaviour

Annex A — UX Review Checklist

- ☐ Can the user tell what is fact, claim, inference, hypothesis and assessment?
- ☐ Can every material analytical object expose its provenance?
- ☐ Does the UI distinguish DIRECT / DOCUMENTED / RECONSTRUCTED / HYPOTHETICAL without color alone?
- ☐ Does a denied user learn more than they should?
- ☐ Can a reviewer identify contradictions and gaps?
- ☐ Can a high-impact action be reversed or superseded where appropriate?
- ☐ Does export show exactly what leaves the system?
- ☐ Are unknown and zero/none clearly different?

- ☐ Are error and partial states explicit?
- ☐ Can the workflow be completed with keyboard?
- ☐ Does any styling imply guilt or suspicion without analytical basis?
- ☐ Could AI output be mistaken for evidence or verified fact?

Annex B — Recommended UX Deliverables

Deliverable	Purpose
Critical task-flow diagrams	Model interaction sequences without deciding final IA.
Low-fidelity wireframes	Validate interaction and state behaviour.
State matrices	Document empty/loading/error/permission/conflict states.
Interaction prototypes	Test evidence, merge, value-flow, hypothesis, review and export flows.
Design-system semantic tokens	Standardize status, caution, provenance and uncertainty behaviour.
Accessibility annotations	Specify keyboard/focus/alternative representations.
Usability test scripts	Validate required UT-01..UT-08 scenarios.
UX decision log	Record consequential deviations/decisions and handoffs to IA.